

Representation of column vectors

basis
synthesize / linear
analyze independence
&
orthogonality

Let us revisit the representation of a column vector of length N . Let

$$\mathbf{x} = \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix} \in \mathbb{C}^N.$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

We also define $\mathbf{e}_0, \mathbf{e}_1, \dots, \mathbf{e}_{N-1} \in \mathbb{C}^N$ by

$$\mathbf{e}_0 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{e}_1 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots \quad \mathbf{e}_{N-1} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

standard basis
vectors

Here we use the zero-based index system. This means that the top entry is the 0th entry. (The standard index system starts with the first entry. We intentionally use the zero-based index system for a later use, which is consistent with discrete-time signals of infinite length.) In other words, for each $n = 0, 1, \dots, N-1$, the vector $\mathbf{e}_n$ satisfies that its n th entry is 1 and the other entries are zero.

We are interested in the following question: How can we write $\mathbf{x}$ as a linear combination of $\mathbf{e}_0, \mathbf{e}_1, \dots, \mathbf{e}_{N-1}$? This means that we want to find some constant $a_0, a_1, \dots, a_N \in \mathbb{C}$ such that

Synthesis: represent $\vec{x}$ as
a linear combination
of $\vec{e}_0, \dots, \vec{e}_{N-1}$

$$\mathbf{x} = \sum_{n=0}^{N-1} a_n \mathbf{e}_n = a_0 \mathbf{e}_0 + a_1 \mathbf{e}_1 + \dots + a_{N-1} \mathbf{e}_{N-1}.$$

The right-hand side is called a linear combination of $\mathbf{e}_0, \mathbf{e}_1, \dots, \mathbf{e}_{N-1}$ and the constants $a_0, a_2, \dots, a_{N-1}$ are the weights or coefficients for the linear combination.

Indeed, this is rather a trivial question and we already know the answer. The weights are simply given by

$$a_0 = x[0], a_1 = x[1], \dots, a_{N-1} = x[N-1].$$

It is straightforward to verify that

$$\begin{aligned} \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix} &= \begin{bmatrix} x[0] \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ x[1] \\ \vdots \\ 0 \end{bmatrix} + \dots + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ x[N-1] \end{bmatrix} \\ &= x[0] \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + x[1] \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + \dots + x[N-1] \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}. \end{aligned}$$

$\underbrace{\quad\quad\quad}_{\mathbf{e}_0} \quad \underbrace{\quad\quad\quad}_{\mathbf{e}_1} \quad \dots \quad \underbrace{\quad\quad\quad}_{\mathbf{e}_{N-1}}$

This example illustrates a special case of “orthonormal basis”. This is what we want to review today.

Basis

$$= \begin{bmatrix} \vec{e}_0 & \vec{e}_1 & \dots & \vec{e}_{N-1} \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}$$

$\underbrace{\hspace{10em}}_{\mathbf{I}_N}$

A collection of N column vectors of length N , $\{\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}\}$, is called a “basis” for $\mathbb{C}^N$ if any vector $\mathbf{x}$ of length N is written as a linear combination of $\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1} \in \mathbb{C}^N$. (The definition remains valid as we replace all complex numbers by real numbers, that is, we replace $\mathbb{C}^N$ by $\mathbb{R}^N$. We consider complex valued vectors for its connection to the discrete Fourier transform.)

We need to recall the notion of linear independence. We say that vectors $\{\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}\}$ are “linearly independent” if

$$\sum_{n=0}^{N-1} a_n \mathbf{f}_n = \mathbf{0} \implies a_n = 0, \quad \forall n = 0, \dots, N-1. \quad (1)$$

$$\begin{bmatrix} \vec{f}_0 & \vec{f}_1 & \dots & \vec{f}_{N-1} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \end{bmatrix} = \vec{0}$$

$\underbrace{\hspace{10em}}_{N \times N} \quad \underbrace{\hspace{10em}}_{N \times 1}$

In other words, any $\mathbf{f}_n$ cannot be expressed as a linear combination of the other vectors. Let us define an N -by- N matrix $\mathbf{F}$ by concatenating $\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}$ as follows:

$$\mathbf{F} = \begin{bmatrix} \mathbf{f}_0 & \mathbf{f}_1 & \dots & \mathbf{f}_{N-1} \end{bmatrix}.$$

Another goal we pursue here is to get familiar with the compact linear algebra notations. Let

$$\mathbf{a} = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \end{bmatrix} \in \mathbb{C}^N.$$

Then the linear combination of vectors $\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}$ with weights $a_0, a_1, \dots, a_{N-1}$ is compactly written as a matrix-vector multiplication as

$$\sum_{n=0}^{N-1} a_n \mathbf{f}_n = \mathbf{F} \mathbf{a}.$$

These are well-known facts from linear algebra.

1. There are at most N linearly independent vectors of length N .
2. If $\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}$ are linearly independent, then the matrix $\mathbf{F}$ is invertible. (There exists a matrix $\mathbf{F}^{-1}$ such that $\mathbf{F} \mathbf{F}^{-1} = \mathbf{F}^{-1} \mathbf{F} = \mathbf{I}_N$, where $\mathbf{I}_N$ denotes the N -by- N identity matrix.)

Let us connect these facts to a notion of basis. We claim that $\{\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}\}$ provides a basis for $\mathbb{C}^N$ if $\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}$ are linearly independent. In other words, every $\mathbf{x}$ is written as a linear combination of $\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}$. We can show this by contradiction. Assume that there exists a vector $\mathbf{x}$ which is not written as a linear combination of $\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}$. Then we should have $\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}$ and $\mathbf{x}$ are linearly independent. However, by the first fact above, the max number of linearly independent vectors is N . Therefore it is a contradiction.

There is one more important observation. The weights $a_0, a_1, \dots, a_{N-1}$ in the “representation” of $\mathbf{x}$ as a linear combination of $\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}$ are uniquely determined. First the statement $\mathbf{x}$ is a linear combination of vectors $\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}$ with weights $a_0, a_1, \dots, a_{N-1}$ is rewritten as

$$\mathbf{x} = \mathbf{F}\mathbf{a}.$$

Since $\mathbf{F}$ is invertible, then $\mathbf{a}$ is given by

$$\mathbf{F}^{-1}\mathbf{x} = \mathbf{F}^{-1}\mathbf{F}\mathbf{a} = \mathbf{a}.$$

Let us introduce two important keywords here. The operation that recovers vector $\mathbf{x}$ from the coefficient vector $\mathbf{a}$ by

$$\mathbf{x} = \mathbf{F}\mathbf{a}$$

Synthesis

Any $\vec{x}$ can be written as $\mathbf{F}\vec{a}$

$$\circ R$$

$$\sum_{k=0}^{N-1} a_k \vec{f}_k$$

is called “synthesis”. The operation that computes the coefficient vector $\mathbf{a}$ for $\mathbf{x}$ by

$$\mathbf{a} = \mathbf{F}^{-1}\mathbf{x}$$

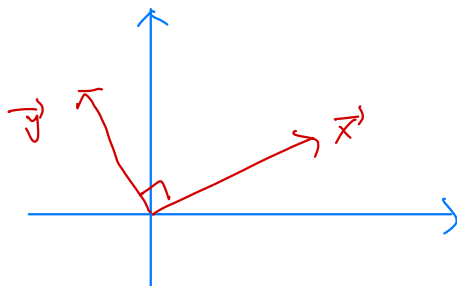
analysis

The weights in linear combination can be computed as $\mathbf{F}^{-1}\vec{x}$

is called “analysis”.

Orthogonality

Example 1 (Orthogonality in 2D)



$$\vec{x} \cdot \vec{y} = 0 \quad \Leftrightarrow \quad \vec{x} \text{ and } \vec{y} \text{ are orthogonal}$$

How do we define the orthogonality in N -dimension?

Let

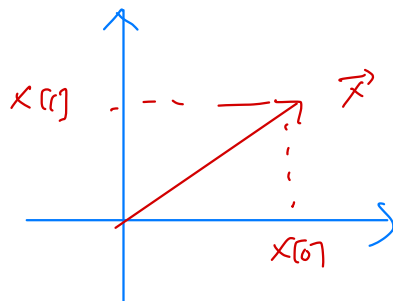
$$\mathbf{x} = \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} y[0] \\ y[1] \\ \vdots \\ y[N-1] \end{bmatrix} \in \mathbb{C}^N.$$

We say that two vectors $\mathbf{x}$ and $\mathbf{y}$ are orthogonal if

$$\mathbf{x}^H \mathbf{y} = \begin{bmatrix} \overline{x[0]} & \overline{x[1]} & \dots & \overline{x[N-1]} \end{bmatrix} \begin{bmatrix} y[0] \\ y[1] \\ \vdots \\ y[N-1] \end{bmatrix} = \sum_{n=0}^{N-1} \overline{x[n]} y[n] = 0.$$

Length of vectors

Example 2 (length of vectors in 2D)



$$\|\vec{x}\| = \vec{x} \cdot \vec{x} = \sqrt{x[0]^2 + x[1]^2}$$

How do we define the length in N -dimension?

For

$$\mathbf{x} = \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix},$$

we define the Euclidean norm of $\mathbf{x}$ by

$$\|\mathbf{x}\| = \sqrt{\sum_{n=0}^{N-1} |x[n]|^2} = \sqrt{\vec{x} \cdot \vec{x}}$$

The Euclidean norm measures the length of $\mathbf{x}$. We say that $\mathbf{x}$ is normalized (with respect to the Euclidean norm) if

$$\|\mathbf{x}\| = 1.$$

$$\mathbf{F}^H \vec{x} = \begin{bmatrix} \vec{f}_0^H \\ \vec{f}_1^H \\ \vdots \\ \vec{f}_n^H \end{bmatrix} \vec{x} = \begin{bmatrix} \vec{f}_0^H \vec{x} \\ \vec{f}_1^H \vec{x} \\ \vdots \\ \vec{f}_n^H \vec{x} \end{bmatrix}$$

Orthonormal basis

We are interested in a class of bases (plural of basis) which satisfy some special properties. We say that $\{\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}\}$ provide an “orthonormal” basis for $\mathbb{C}^N$ if the following two properties are satisfied.

1. For any $m \neq m'$, $\mathbf{f}_m$ and $\mathbf{f}_{m'}$ are orthogonal.

$$\vec{f}_m \cdot \vec{f}_{m'} = 0 \quad \forall m \neq m'$$

2. For every m , $\|\mathbf{f}_m\| = 1$.

$$\mathbf{F} = [\vec{f}_0 \ \vec{f}_1 \ \dots \ \vec{f}_N] \quad \text{size}$$

In fact, the two properties are sufficient to say that $\{\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}\}$ provide a basis for $\mathbb{C}^N$. It follows because pairwise orthogonal vectors are linearly independent.

How does it help to have the above two properties? To see this, let us compute $\mathbf{F}^H \mathbf{F}$ as follows:

$$\begin{aligned} \mathbf{F}^H \mathbf{F} &= \begin{bmatrix} \mathbf{f}_0^H \\ \mathbf{f}_1^H \\ \vdots \\ \mathbf{f}_{N-1}^H \end{bmatrix} [\mathbf{f}_0 \ \mathbf{f}_1 \ \dots \ \mathbf{f}_{N-1}] \\ &= \begin{bmatrix} \mathbf{f}_0^H \mathbf{f}_0 & \mathbf{f}_0^H \mathbf{f}_1 & \dots & \mathbf{f}_0^H \mathbf{f}_{N-1} \\ \mathbf{f}_1^H \mathbf{f}_0 & \mathbf{f}_1^H \mathbf{f}_1 & \dots & \mathbf{f}_1^H \mathbf{f}_{N-1} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{f}_{N-1}^H \mathbf{f}_0 & \mathbf{f}_{N-1}^H \mathbf{f}_1 & \dots & \mathbf{f}_{N-1}^H \mathbf{f}_{N-1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} = \mathbf{I}_N. \end{aligned}$$

$$\mathbf{F}^H = \begin{bmatrix} \vec{f}_0^H \\ \vdots \\ \vec{f}_N^H \end{bmatrix}$$

Therefore we deduce that $\mathbf{F}^H = \mathbf{F}^{-1}$. This fact is very helpful particularly for the analysis procedure. Instead of applying the inverse of matrix $\mathbf{F}$, which is often quite expensive to compute, we can only apply the Hermitian transpose of $\mathbf{F}$.

analysis

$$\text{If } \vec{x} = \mathbf{F} \vec{a}, \text{ then } \vec{a} = \mathbf{F}^{-1} \vec{x}$$

$$\text{If } \mathbf{F}^H \mathbf{F} = \mathbf{I}_N, \text{ then } \vec{a} = \mathbf{F}^H \vec{x}$$

Standard basis

We started with the vectors $\mathbf{e}_0, \mathbf{e}_1, \dots, \mathbf{e}_{N-1}$. As we concatenate them, we obtain the identity matrix, that is

$$\mathbf{E} = [\mathbf{e}_0 \ \mathbf{e}_1 \ \dots \ \mathbf{e}_{N-1}] = \mathbf{I}_N.$$

It is easy to verify that

$$\mathbf{E}^H \mathbf{E} = \mathbf{I}_N.$$

Therefore, $\{\mathbf{e}_0, \mathbf{e}_1, \dots, \mathbf{e}_{N-1}\}$ provides an orthonormal basis for $\mathbb{C}^N$. In fact, we call this the standard basis or canonical basis.

Fourier basis

Let $\mathbf{f}_n$ be defined by

$$\mathbf{f}_n = \frac{1}{\sqrt{N}} \begin{bmatrix} (e^{j2\pi/N})^{0 \cdot n} \\ (e^{j2\pi/N})^{1 \cdot n} \\ \vdots \\ (e^{j2\pi/N})^{(N-1)n} \end{bmatrix}, \quad n = 0, \dots, N-1.$$

Can we show that $\{\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{N-1}\}$ provides an orthonormal basis for $\mathbb{C}^N$?

First we show that

$$\mathbf{f}_n^H \mathbf{f}_n = \frac{1}{N} \sum_{k=0}^{N-1} (e^{-j2\pi/N})^{kn} (e^{j2\pi/N})^{kn} = \frac{1}{N} \sum_{k=0}^{N-1} 1 = 1.$$

Let $n' \neq n$. Then we also have

$$\begin{aligned}
 \mathbf{f}_{n'}^H \mathbf{f}_n &= \frac{1}{N} \sum_{k=0}^{N-1} \left(e^{-j2\pi/N} \right)^{kn'} \left(e^{j2\pi/N} \right)^{kn} \\
 &= \frac{1}{N} \sum_{k=0}^{N-1} \left(e^{j2\pi/N} \right)^{k(n-n')} \\
 &= \frac{1}{N} \cdot \frac{1 - \left(e^{j2\pi/N} \right)^{N(n-n')}}{1 - \left(e^{j2\pi/N} \right)^{n-n'}} \\
 &= \frac{1}{N} \cdot \frac{1 - \left(e^{j2\pi} \right)^{n-n'}}{1 - \left(e^{j2\pi/N} \right)^{n-n'}} = 0.
 \end{aligned}$$

$e^{j\frac{2\pi}{P}kt}$ is defined for all $k \in \mathbb{Z}$ $\lim_{N \rightarrow \infty} \underbrace{\sum_{k=-N}^N x_k e^{j\frac{2\pi}{P}kt}}_{\text{order } N \text{ approximation}} = x(t)$

Fourier Series:

- A periodic signal, with period P , satisfies

$$x(t) = x(t + nP) \text{ for all integers } n.$$

Examples: sine wave, square wave, triangle wave, ...

We consider the vector space of all periodic signals with period P .

- The Fourier series (FS) says that a P -periodic signal $x(t)$ can be written as an infinite sum of complex exponentials:

Complex sinusoid of freq. $\frac{2\pi}{P}k \rightarrow P$ -periodic

$$x(t) = \sum_{k=-\infty}^{\infty} x_k e^{j\frac{2\pi}{P}kt}, \quad x_k \triangleq \frac{1}{P} \int_0^P x(t) e^{-j\frac{2\pi}{P}kt} dt,$$

where $\{x_k\}_{k \in \mathbb{Z}}$ are known as the FS coefficients.

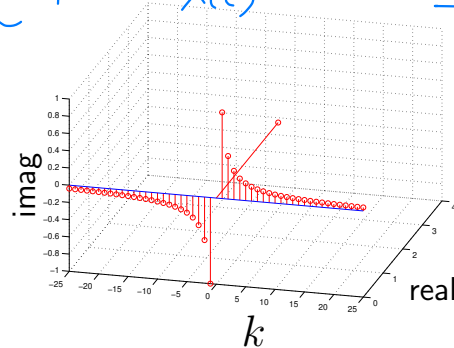
$e^{j\frac{2\pi}{P}k(t+P)} = e^{j\frac{2\pi}{P}kt} e^{j2\pi k}$

$e^{j\frac{2\pi}{P}kt} \cdot x(t)$

- Example — sawtooth wave:

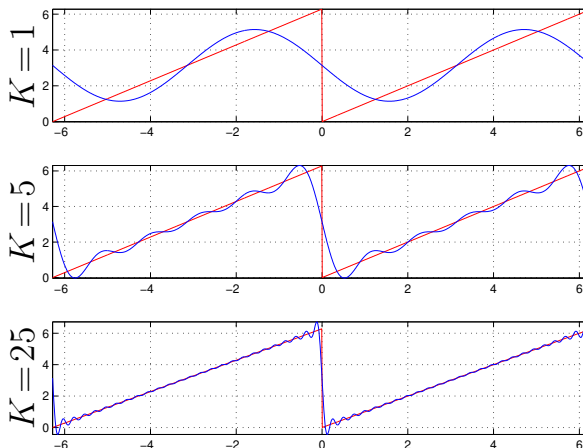
$$x(t) = t \text{ for } t \in [0, P), P = 2\pi$$

$$x_k = \frac{j}{k} |_{k \neq 0}, \quad x_0 = \frac{P}{2}$$



K -truncated FS

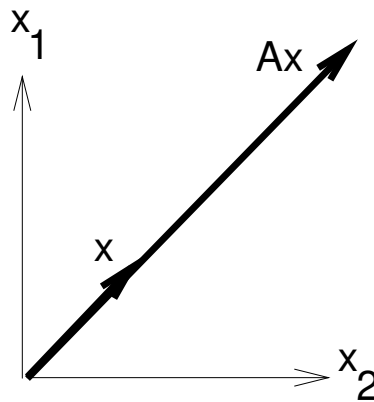
$$\sum_{k=-K}^K x_k e^{j\frac{2\pi}{P}kt}$$



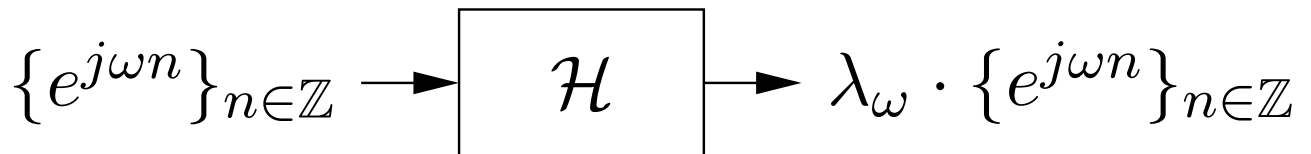
Eigenfunctions of DT LTI Systems:

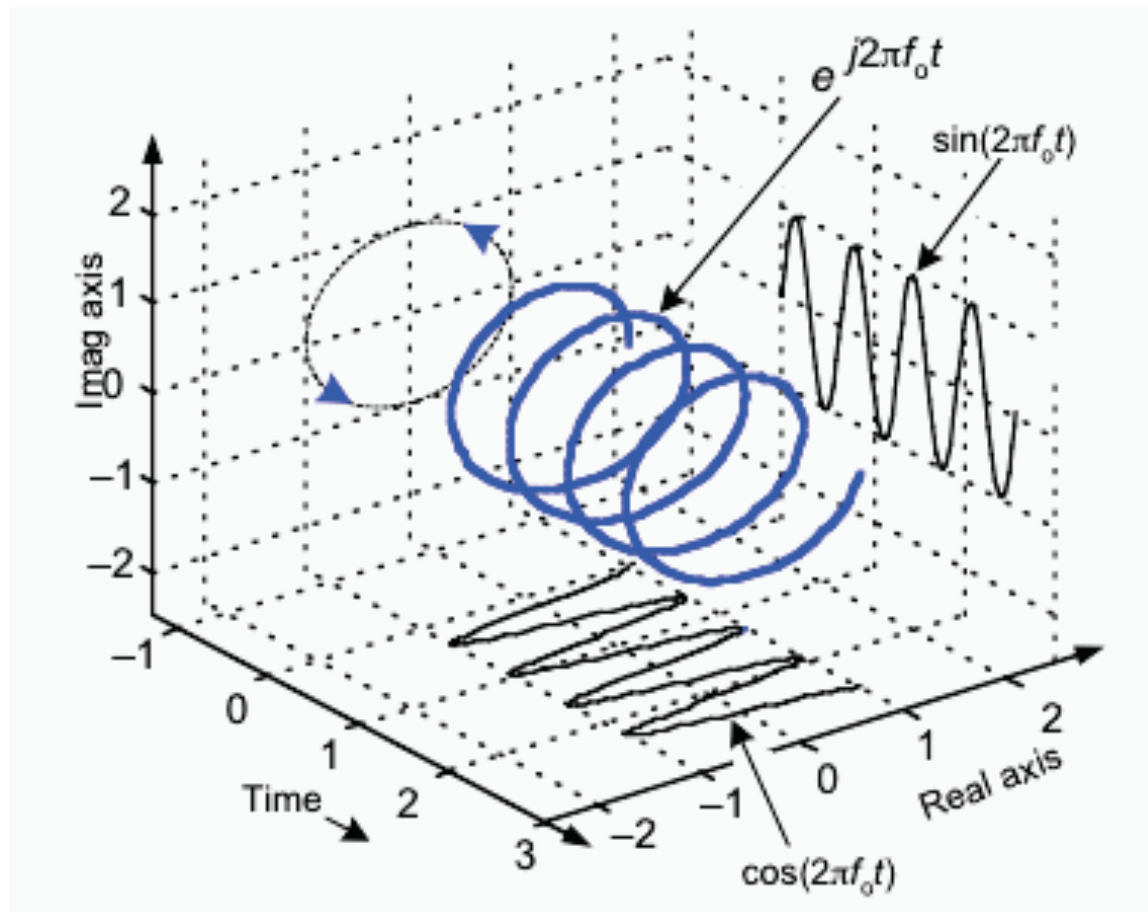
- An analogy:

$$A\mathbf{x} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



- LTI System:

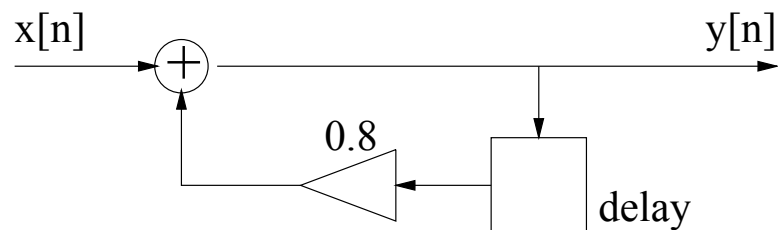




IIR Example:

Consider a recursive LTI system:

$$y[n] = x[n] + 0.8y[n - 1]$$



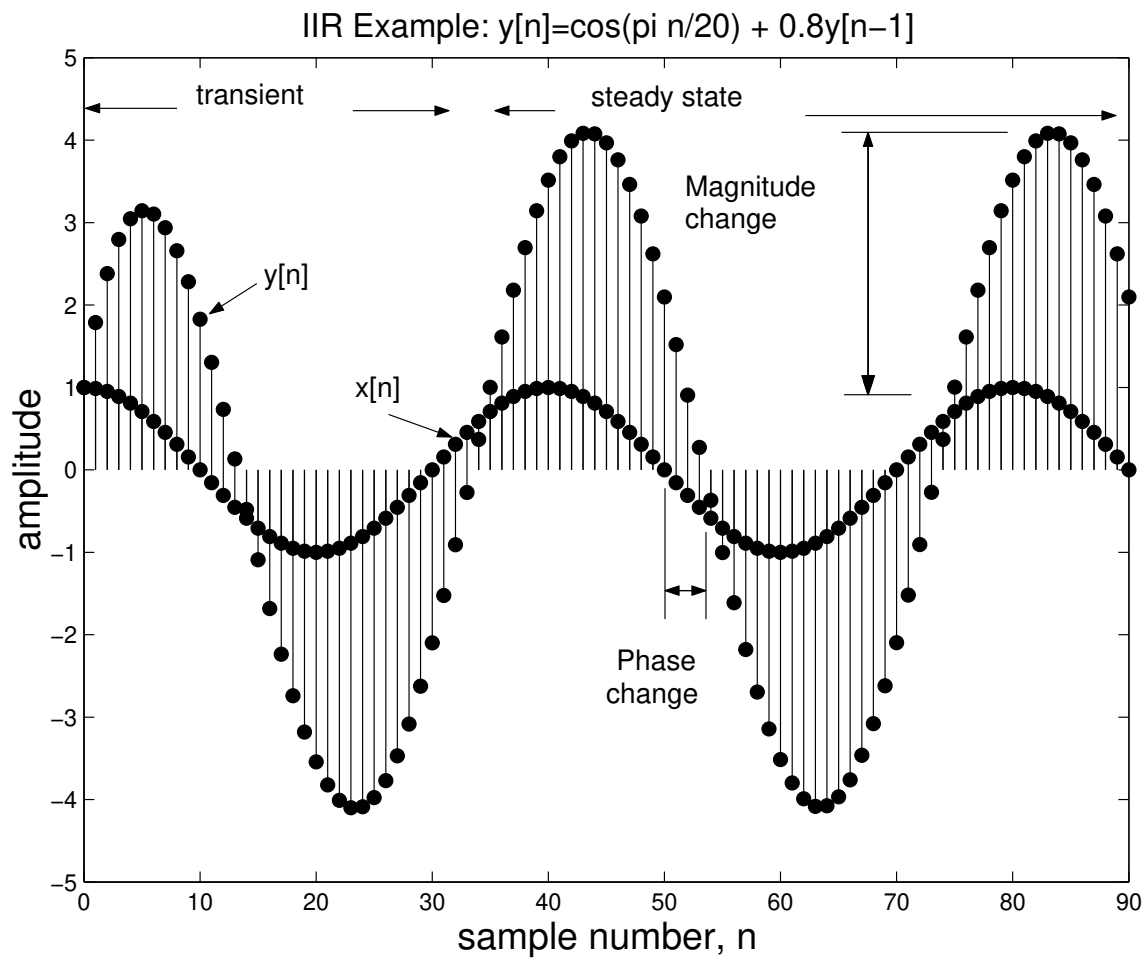
$$x[n] = \cos(\pi n/20)u[n]$$

$$|H(e^{j\omega})| = 4.093 \quad \text{for } \omega = \pi/20$$

$$\angle H(e^{j\omega}) = -0.538 \text{ radians} \quad \text{for } \omega = \pi/20$$

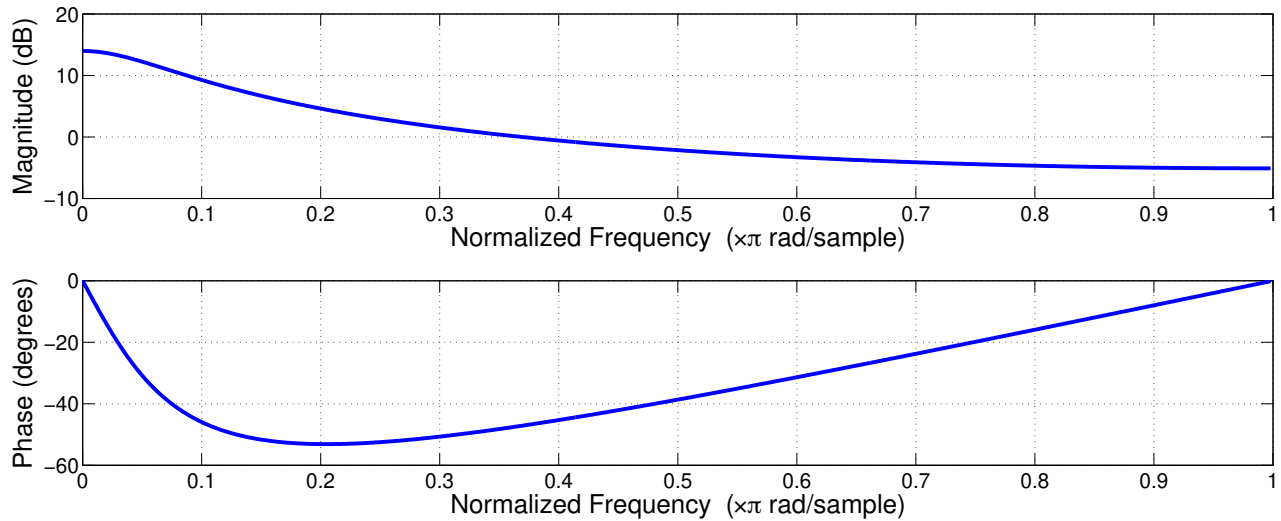
$$y_{ss}[n] = 4.093 \cos(\pi n/20 - 0.538)$$

IIR Example, continued:



IIR Example, continued:

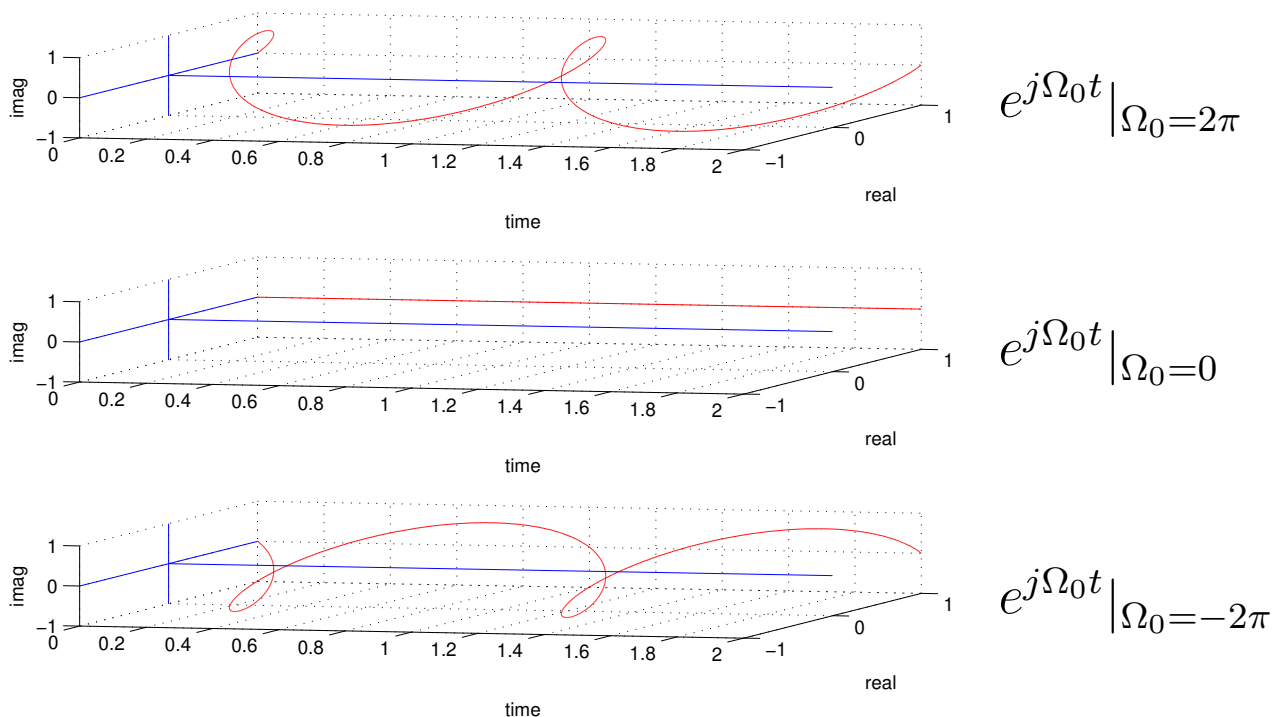
Frequency response plots (magnitude and phase)



```
>> freqz([1],[1 -0.8])
```

Negative frequencies:

- The CTFT allows negative frequencies. What are they?
- Recall that a “pure” tone at Ω_0 radians/sec can be written as $e^{j\Omega_0 t} \xleftrightarrow{\text{CTFT}} 2\pi\delta(\Omega - \Omega_0)$.
- The signal $e^{j\Omega_0 t}$ is a corkscrew: for $\Omega_0 > 0$, it spins counter-clockwise and, for $\Omega_0 < 0$, it spins clockwise.



- In essence, the CTFT implies that any signal can be constructed as a sum of (possibly infinitely many) scaled corkscrews, each with a distinct frequency. The sign of the frequency simply denotes the direction of spin.

- What if the signal is a real-valued sinusoid?

$$\begin{aligned}\cos(\Omega_0 t + \theta) &= \frac{1}{2} \left(e^{j(\Omega_0 t + \theta)} + e^{-j(\Omega_0 t + \theta)} \right) \\ &= \frac{e^{j\theta}}{2} e^{j\Omega_0 t} + \frac{e^{-j\theta}}{2} e^{-j\Omega_0 t}\end{aligned}$$

Thus it contains both positive and negative frequencies.

- How do we generate/represent complex signals in hardware?

$$e^{j(\Omega_0 t + \theta)} = \cos(\Omega_0 t + \theta) + j \sin(\Omega_0 t + \theta)$$

Using two real-valued wires/buses.

FIR Example:

$$y[n] = 0.5x[n] + x[n - 1] + 0.5x[n - 2]$$

This is a second-order FIR filter, and hence is a linear time-invariant system. We have that

$$h[n] = 0.5\delta[n] + \delta[n - 1] + 0.5\delta[n - 2]$$

To find $H(e^{j\omega})$, we start with the definition

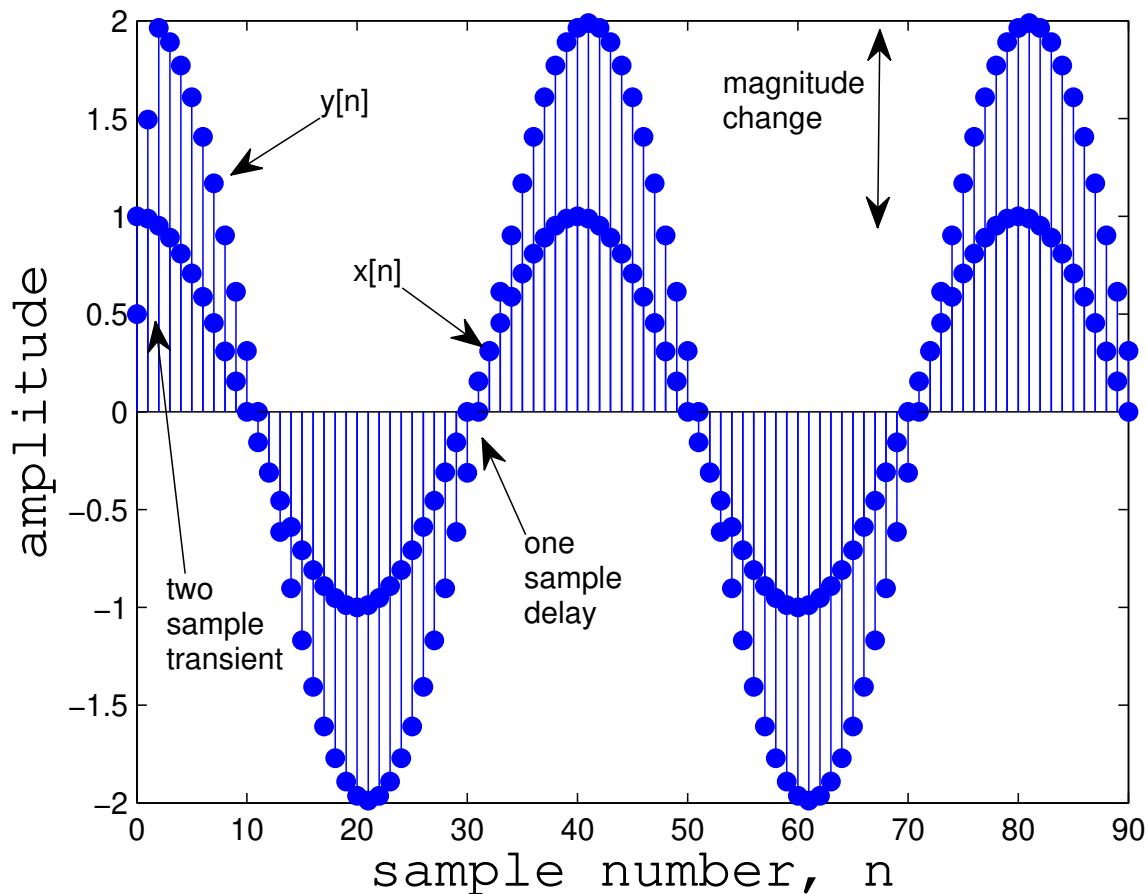
$$\begin{aligned} H(e^{j\omega}) &= \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \\ &= \frac{1}{2}e^0 + 1e^{-j\omega} + \frac{1}{2}e^{-j\omega 2} \\ &= e^{-j\omega} \left\{ \frac{1}{2}e^{j\omega} + 1 + \frac{1}{2}e^{-j\omega} \right\} \\ &= e^{-j\omega} \{1 + \cos(\omega)\} \end{aligned}$$

Thus, $|H(e^{j\omega})| = 1 + \cos(\omega)$ and $\angle H(e^{j\omega}) = -\omega$.

FIR Example, continued:

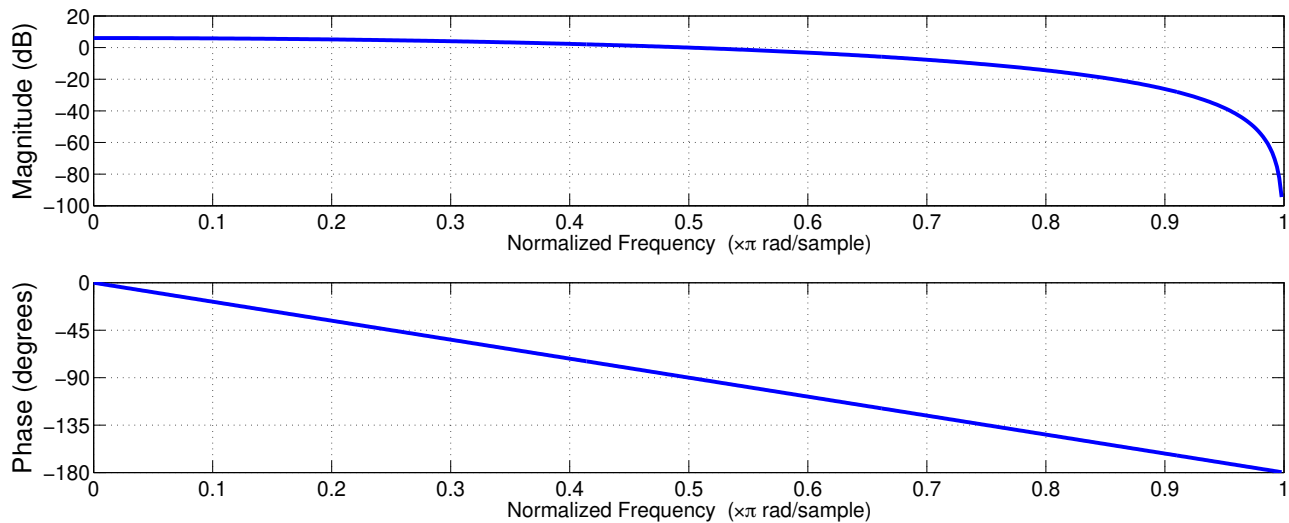
For $x[n] = \cos(\pi n/20)u[n]$ we have $A = 1$, $\omega = \pi/20$ and $\phi = 0$. Thus,

$$\begin{aligned}y_{ss}[n] &= A |H(e^{j\omega})| \cos(\omega n + \phi + \angle H(e^{j\omega})) \\&= 1 \cdot |H(e^{j\pi/20})| \cos(\pi n/20 + 0 + \angle H(e^{j\pi/20})) \\&= 1.988 \cos(\pi n/20 - \pi/20)\end{aligned}$$



FIR Example, continued:

Frequency response plots (magnitude and phase)



```
>> b = [0.5, 1, 0.5];  
>> a=1;  
>> freqz(b,a);
```

Eigensignals of CT-LTI systems:

- Consider complex *sinusoidal* input $x(t) = e^{j\Omega_0 t}$, $\Omega_0 \in \mathbb{R}$.

For LTI system $\mathcal{H}_c$, we have

$$\begin{aligned}\mathcal{H}_c\{x(t)\} &= \int h(\tau)x(t-\tau)d\tau = \int h(\tau)e^{j\Omega_0(t-\tau)}d\tau \\ &= e^{j\Omega_0 t} \underbrace{\int h(\tau)e^{-j\Omega_0\tau}d\tau}_{\triangleq H_c(j\Omega_0)} = x(t) \underbrace{H_c(j\Omega_0)}_{\text{scaling factor}},\end{aligned}$$

using Laplace-transform notation (more on this later...).

The output is simply a scaled version of the input!

- A signal which passes through system $\mathcal{H}_c$ with only a gain/phase change is an “*eigensignal*” of $\mathcal{H}_c$, and the corresponding scaling factor is its “*eigenvalue*.”

Sinusoids are eigensignals of LTI systems.

- This property yields an important implication for any *P-periodic signal* $x(t)$. Notice that if we expand such a signal via its *Fourier series* (FS) as

$$x(t) = \sum_{k=-\infty}^{\infty} x_k e^{j\frac{2\pi}{P}kt}$$

then the linearity of $\mathcal{H}_c$ implies

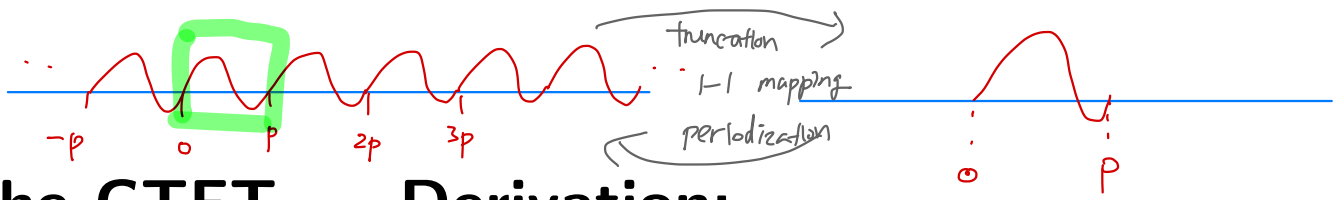
$$\mathcal{H}_c\{x(t)\} = \sum_{k=-\infty}^{\infty} \underbrace{x_k H_c(j\frac{2\pi k}{P})}_{y_k} \overbrace{e^{j\frac{2\pi}{P}kt}}^{h_k}.$$

- Thus, if we represent the input signal by $\{x_k\}_{k \in \mathbb{Z}}$, the output signal by $\{y_k\}_{k \in \mathbb{Z}}$, and the system by $\{h_k\}_{k \in \mathbb{Z}}$, then the operation of passing the signal through the system is simply multiplication of the corresponding coefficients: $y_k = h_k x_k$!

For an LTI system with a periodic input, the output can be computed algebraically.

- Note, however, that the example above applies only to *periodic* signals. Can we extend it to non-periodic signals?

Yes, through the continuous-time Fourier transform (CTFT)...

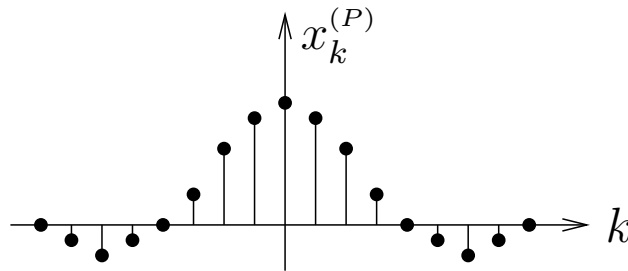


The CTFT — Derivation:

- We now derive the *continuous-time Fourier transform* (CTFT) as the $P \rightarrow \infty$ limit of the Fourier series (FS).
- Consider an arbitrary P -periodic signal $x^{(P)}(t)$ with corresponding FS coefficients $\{x_k^{(P)}\}_{k \in \mathbb{Z}}$, where

$$x_k^{(P)} = \frac{1}{P} \int_0^P x^{(P)}(t) e^{-j \frac{2\pi}{P} kt} dt = \frac{1}{P} \int_{-P/2}^{P/2} x^{(P)}(t) e^{-j \frac{2\pi}{P} kt} dt.$$

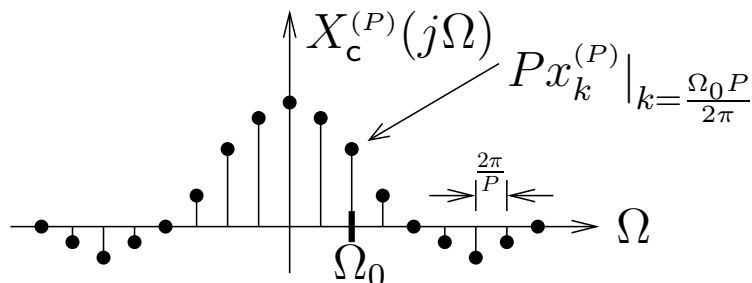
- The collection $\{x_k^{(P)}\}_{k \in \mathbb{Z}}$ constitutes the FS “spectrum”:

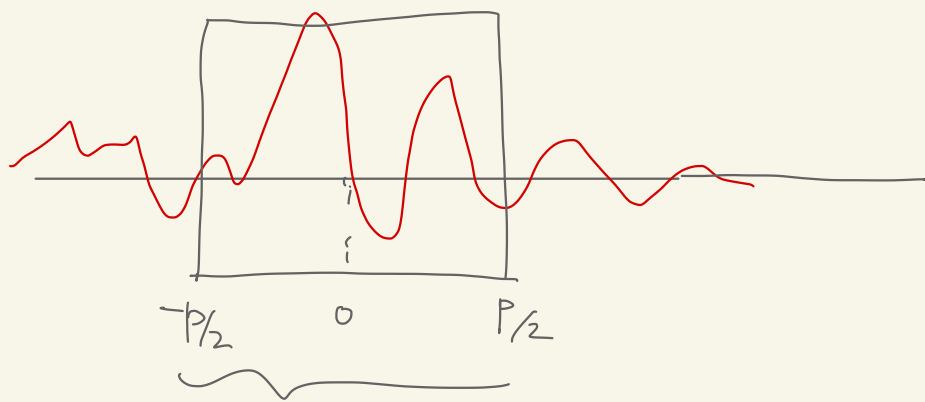


- Now define the “frequency” $\Omega \triangleq \frac{2\pi k}{P}$ (in rad/sec) and, from it, the (scaled) spectrum

$$X_c^{(P)}(j\Omega) \triangleq P x_k^{(P)} = \int_{-P/2}^{P/2} x^{(P)}(t) e^{-j\Omega t} dt.$$

- We can visualize the spectrum $X_c^{(P)}(j\Omega)$ as follows:





truncation



periodization

